

## Data Science Paper Guide

### Player Profile

**Name:** King Dheniel N. Conception

**Role:** 4th Year Computer Science Student / Gaming Enthusiast

**Background:** As a student specializing in Machine Learning with a deep interest in data science, King maintains a large digital gaming library across multiple platforms (PC, PlayStation, Switch). He is curious about how his gaming habits translate into data and wants to use statistical methods to prove that personal digital behavior follows predictable scientific patterns.

**Goals:** To transform raw gaming logs into actionable insights and establish a robust framework for personal data analysis that can be replicated by other researchers.

### Problem Statement

Digital gaming libraries accumulate vast amounts of personal data, yet these records often remain unexamined at a personal level. Without a structured analytical approach, it is difficult to identify core engagement drivers or understand how session frequency impacts total time investment in a highly skewed data environment. This study addresses this gap by applying a quantitative statistical framework to model personal gaming behavior and demonstrate that digital consumption follows predictable, evidence-based patterns.

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### Research Questions

**General Research Question:** How can a quantitative statistical framework be utilized to analyze and model the engagement patterns and behavioral trends within a personal digital gaming library?

#### Specific Research Questions:

1. What is the distribution profile of the gaming library regarding Time Played Hours and Play Count, and how does it show the difference between high-engagement titles from the rest of the collection?
2. Is there a statistically significant relationship between session frequency & total duration and what is the strength of this association?
3. How do gaming preferences evolve across different platforms and categories, and which genres currently lead engagement compared to historical data?

## Objectives of the Study

**General Objective:** To develop and apply a quantitative statistical framework for analyzing and modeling player engagement patterns and behavioral trends within a personal digital gaming library.

### Specific Objectives:

1. To characterize the distribution profile of the gaming library by calculating descriptive statistics for Time Played Hours and Play Count, identifying the presence of power-law patterns and high-engagement "outlier" titles.
  2. To evaluate the statistical relationship between session frequency and total play duration by applying Spearman's Rank Correlation to determine the significance and strength of the association.
  3. To analyze the temporal evolution of gaming preferences by comparing historical genre and platform data against recent engagement metrics to identify shifts in player interest.
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### Objective 1: Characterizing the Distribution Profile

*This objective focuses on defining the "shape" of your data and identifying high-engagement outlier titles.*

- **Table 2: Descriptive Statistics Summary** – Presents the mean, median, standard deviation, skewness, and kurtosis for both *Time Played Hours* and *Play Count*.
  - **Figure 1: Histogram of Time Played Hours** – Provides visual evidence of the right-skewed distribution, where most games cluster at low playtime values.
  - **Figure 2: Histogram of Play Count** – Illustrates the distribution of session frequency, which also follows a power-law pattern with a small number of games showing high session counts.
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### Objective 2: Evaluating Statistical Relationships

*This objective aims to prove the relationship between session frequency (Play Count) and total duration (Time Played Hours).*

- **Figure 5: Correlation Matrix Heatmap** – A color-coded visualization displaying the strength of the relationship between engagement metrics.
  - **Table 3: Summary Comparative Analysis** – Reports the Pearson  $r$  and Spearman  $\rho$  (0.7942) results, including the statistically significant p-value ( $<0.001$ ).
  - **Figure 6: Scatter Plot of Play Count vs. Time Played Hours** – Shows the upward trend line and explicitly identifies outliers such as *Jurassic World Evolution 2* and *Grand Theft Auto V*.
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Objective 3: Analyzing Temporal and Category Evolution

These objective maps the shift in player preferences across different platforms and genres from 2021 to 2026.

- **Figure 3: Games Added Per Year (2021–2026)** – A bar chart tracking the growth rate of the digital gaming library.
- **Figure 4: Total Playtime Per Year (2021–2026)** – A line chart displaying the progression of gaming engagement, peaking significantly in 2025.
- **Figure 7: Playtime Distribution by Platform Box Plot** – Compares engagement levels across various hardware environments, such as PC and consoles.
- **Table 4: Category Analysis Summary** – Provides a detailed breakdown of engagement metrics for each standardized genre (Simulation, Action, Shooter, etc.).
- **Figure 8: Top Category Per Year (2021–2026)** – Visualizes the historical shift in genre dominance over the five-year study period.
- **Figure 9: Playtime by Category (Past 30 Days)** – Identifies current gaming interests (Shooter) compared to long-term historical trends.

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Summary and Interpretations

These tables consolidate all statistical findings into actionable insights.

- **Table 5: Summary of Statistical Analysis** – The "master table" integrating sample size, descriptive metrics, and correlation results for quick reference.
- **Table 6: Summary of Behavioral Interpretations and Statistical Key Findings** – Translates quantitative data into practical insights regarding player habits and behavior modeling.

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Executive Summary: Analyzing Gaming Engagement Patterns

This study successfully established a Statistical Framework to analyze player behavior over a five-year period (2021–2026) using data from 72 active games. By utilizing Descriptive Statistics and Spearman’s Rank Correlation, the research proves that personal digital behavior follows predictable and evidence-based patterns.

Key Findings:

- **Selective Engagement Profile:** The study identified that the gaming library is "highly right-skewed" (Skewness: 4.90), meaning a few "high-engagement titles" such as *Jurassic World Evolution 2* and *Grand Theft Auto V* account for the majority of total playtime.

- **Strong Behavioral Correlation:** The framework produced a strong Spearman Correlation of 0.7942, confirming that session frequency (Play Count) is a critical and reliable indicator of the total time (Time Played) invested in a game.
- **Genre Evolution:** Temporal analysis revealed a significant shift in player interest; while Simulation (824.2 hours) and Action (764.0 hours) were historically dominant, the Shooter genre (125.6 hours) led engagement in the final 30 days of the study.
- **Peak Engagement Patterns:** The year 2025 was identified as the period of highest activity (1,683 hours), which is attributed to external factors such as academic schedules and specific game releases.

**Conclusion and Implications:** The study achieved its goal of modeling engagement patterns using non-parametric tools. This framework not only provides self-insight for better time management but also serves as a scientific baseline for understanding how technological commitment fluctuates over time.